



Bornean Rain Forest

A threatened habitat governed by three nations, home to more than 15,000 plant species and one of the oldest and most biodiverse rain forests on Earth

Shared by Malaysia, Brunei, and Indonesia, the island of Borneo is home to Asia's largest rain forest, which is also one of the world's most ancient at 130 million years old—around 70 million years older than the Amazon Rain Forest (see pp.120–21). Borneo's rain forest is hugely biodiverse: while the island amounts to just 1 percent of Earth's landmass, its rain forest contains around 6 percent of the planet's plant and animal species.

A wealth of hardwood

The plants include a family of tropical hardwood trees called dipterocarps, many of which reach heights of 200 ft (60 m). Most of the 600-plus species of dipterocarps grow in Southeast Asia at elevations up to 3,300 ft (1,000 m), and Borneo's lowland rain forests contain more of these trees than any other location. The island's 270 species of dipterocarps include the highly prized Borneo ironwood, whose wood is so dense it never needs treating. Borneo's rain forests, including those covering its more mountainous interior, are also rich in many other types of life. More than 360 new plant species have been discovered here



◁ CALLING CARD

The nocturnal tree-dwelling tokay gecko is just one of many reptiles in Borneo and is named for its "to-kay!" call. It grows up to 16 in (40 cm) long.

since 1995, and the island is home to more than 1,400 species of amphibians, mammals, birds, reptiles, and fish—many found nowhere else.

Rain forest under threat

However, this species treasury is in peril. Until the early 1970s, more than three-quarters of Borneo's 287,000 square miles (743,330 square km) were covered in thick tropical rain forest that was densest in the lowlands. Since then, at least one-third has been destroyed. Fires and the planting of oil palm plantations are partly to blame, but it was demand for the highly valuable dipterocarps that resulted in logging on an industrial scale, particularly in the Malaysian states of Sabah and Sarawak in the north of the island, where an estimated 80 percent of rain forest has been lost. To preserve this habitat, in 2007, 85,000 square miles (220,000 square km) of rain forest in the center of the island were designated as a protected area, known as the Heart of Borneo.

Up to **240 tree species** can be found in just **2½ acres (1 hectare)** of **Borneo's rain forest**

△ PROTECTED TREES

The crowns of the tallest trees, known as emergents, rise above the morning mist that covers the rest of the rain forest in the Danum Valley, a conservation area of lowland dipterocarp forest in Sabah.

▷ LIFE SUPPORT

Borneo's rain forests support many highly endangered animals, including the critically endangered Bornean orangutan, which depends on dipterocarp trees for food and shelter.

HOW A STRANGLER FIG GROWS

In most cases, a sticky strangler fig seed germinates high on a branch of a rain forest tree, where it was left by a monkey, bird, or bat that had eaten the fruit. The seedling sends long roots down the trunk of its host to enter the soil. Eventually, a scaffolding of roots encases the trunk, and the host's root system must compete with the strangler fig's. The fig's thick foliage also shades the host tree's crown. Ultimately the host dies, leaving the fig tree behind.

